

Power Systems.

How much time does it take to break even? Considering that 1 kW produces 120 kWh of energy, total yearly savings from a 3 kW solar plant, assuming an average price of ₹7 per unit, amount to ₹30,240. Additionally, factors such as increasing grid electricity tariffs, availability of financial incentives, and net metering also influence the total long-term savings. However, it is important to note that a 3 kW solar system costs around ₹2 lakh without subsidy. After accounting for subsidy, the cost reduces by 39%—to approximately ₹1.2 lakh. Hence, it typically takes around four years to achieve the break-even point before benefits from solar energy start flowing in (See chart ‘Pocket Pinch’).

For those who want to instal RTS but are hindered by the high initial costs, there are many banks and fintechs registered on the scheme’s website that provide financing options. “For an average household requiring a 3-5 kW installation, the cost can range from ₹1.75 lakh to ₹3.5 lakh. This expense can be managed through subsidy and manageable EMIs of around ₹4,000-5,000 per month, equivalent to monthly electricity bills, thereby alleviating the financial burden, advise industry watchers and stakeholders.

“We aim to keep the EMI equal to electricity bills, ensuring no additional burden on the consumer,” says Agarwal. Credit Fair offers no-cost EMIs where the interest cost is borne by a vendor. Regular EMIs are offered at a flat annual interest rate of 9-10%.

SUBSIDY SUPPORT

If you plan to instal rooftop solar panels for a residential household, you can apply for a subsidy un-

POCKET PINCH

- Setting up a 3 kW connection costs around ₹2 lakh
- There is a subsidy of ₹78,000, which brings cost down by 39%

Description	Quantity	Rate	Amount
Solar PV Module (Mono Perc Half Cut-DCR)*	6	15,120	90,720
Solar inverter	1	30,745	30,745
Galvanised iron structure	1	18,000	18,000
ACDB**, DCDB**, SPD, AC wire, DC Wire, connectors, ties, nuts and bolts, earthing	1	15,000	15,000
Net metering	1	15,000	15,000
Installation	1	9,000	9,000
Total cost			1,78,465
GST @ 12%			20,335.80
Gross total amount			1,98,801

NOTE RATE AND AMOUNT ARE IN ₹; *IN HALF-CUT SOLAR PANELS, CELLS ARE CUT IN TWO HALVES TO INCREASE THE NUMBER OF CELLS; DCR STANDS FOR DOMESTIC CONTENT REQUIREMENT

**ALTERNATING CURRENT/DIRECT CURRENT DISTRIBUTION BOX; SPD IS SURGE PROTECTION DEVICE; THERE IS A 25-YEAR NORMAL WARRANTY ON SOLAR PANELS AND SERVICE OF FIVE YEARS ON SOLAR INVERTER

SOURCE PRAKRUTI SOLAR

der the PM Surya Ghar Muft Bijli Yojana. Under the scheme, there is a subsidy of ₹30,000 per kW up to 2 kW and ₹18,000 per kW for additional capacity up to 3 kW. However, the total subsidy for systems larger than 3 kW is capped at ₹78,000.

A consumer can apply for the subsidy through either discom-tendered projects or the National Portal of India, as stated by Minister of New and Renewable Energy R.K. Singh in a written response to a Lok Sabha query last year.

“On the national portal, consumers have the option to select any vendor and choose the brand and quality/efficiency of the solar equipment. The dis-